

Two dimensional random variables

Definition The function $p(x, y)$ is a joint probability distribution (or) probability mass function of the discrete random variables X and Y if

$$(1) \quad P(x, y) \geq 0 \quad \text{for all } x \in X, y \in Y$$

$$(2) \quad \sum_x \sum_y p(x, y) = 1$$

$$\text{where } p(x, y) = P(X=x, Y=y).$$

Pb Two ballpoint pens are selected at random from a box that contains 3 blue, 2 red and 3 green pens. If X is the number of blue pens selected and Y is the number of red pens selected then find

(a) The joint probability function $p(x, y)$

(b) $P(X + Y \leq 1)$

Soln The possible pair values are $(0, 0)$, $(0, 1)$, $(1, 0)$, $(1, 1)$, $(0, 2)$ and $(2, 0)$

Here $(0, 0)$ means selecting 0 blue, 0 red pen

$(1, 0)$ means selecting 1 blue, 0 red pen

$(1, 1)$ means selecting 1 blue, 1 red pen.

Similarly other cases follows.

$$P(X=0, Y=0) = P(0,0) = \frac{{}^3C_2}{{}^8C_2} = \frac{3}{28}$$

$$P(X=1, Y=0) = P(1,0) = \frac{9}{28}$$

$$P(X=2, Y=0) = P(2,0) = \frac{3}{28}$$

$$P(X=0, Y=1) = P(0,1) = \frac{3}{14}$$

$$P(X=1, Y=1) = P(1,1) = \frac{3}{14}$$

$$P(X=0, Y=2) = P(0,2) = \frac{1}{28}$$

	$P(X,Y)$	X		
		0	1	2
Y	0	$\frac{3}{28}$	$\frac{9}{28}$	$\frac{3}{28}$
	1	$\frac{3}{14}$	$\frac{3}{14}$	0
	2	$\frac{1}{28}$	0	0

$$P(X+Y \leq 1) = P(X=0, Y=0) + P(X=0, Y=1) \\ + P(X=1, Y=0)$$

$$= \frac{3}{28} + \frac{3}{14} + \frac{9}{28} = \frac{9}{4}$$

Joint probability density function

The function $f(x, y)$ is a joint probability density function of the continuous random variables X and Y if

$$(1) f(x, y) \geq 0 \quad \text{for all } x \in X, y \in Y.$$

$$(2) \int_{-\infty}^{\infty} \int_{-\infty}^{\infty} f(x, y) dx dy = 1 \quad \left(\text{provided the integral exists} \right).$$

Pb Let X and Y be two continuous random variables has the density function

$$f(x, y) = \begin{cases} \frac{2}{5}(2x + 3y) & , 0 \leq x \leq 1, 0 \leq y \leq 1 \\ 0 & \text{otherwise} \end{cases}$$

(1) Verify $f(x, y)$ is a probability density function.

(2) Find $P(0 < X < \frac{1}{2}, \frac{1}{4} < Y < \frac{1}{2})$

Soln

clearly $f(x, y) \geq 0$ for all x, y

$$\begin{aligned} \int_{-\infty}^{\infty} \int_{-\infty}^{\infty} f(x, y) dx dy &= \int_0^1 \int_0^1 \frac{2}{5}(2x + 3y) dx dy \\ &= \int_0^1 \frac{2}{5} (x^2 + 3xy) \Big|_0^1 dy \\ &= \int_0^1 \frac{2}{5} (1 + 3y) dy = 1. \end{aligned}$$

Thus f is a probability density function.

Remark $P(a < X < b, c < Y < d) = \int_c^d \int_a^b f(x, y) dx dy$

$$\begin{aligned}
 P(0 < X < \frac{1}{2}, \frac{1}{4} < Y < \frac{1}{2}) &= \int_{\frac{1}{4}}^{\frac{1}{2}} \int_0^{\frac{1}{2}} \frac{2}{5} (2x + 3y) dx dy \\
 &= \int_{\frac{1}{4}}^{\frac{1}{2}} \frac{2}{5} (x^2 + 3xy) \Big|_0^{\frac{1}{2}} dy \\
 &= \int_{\frac{1}{4}}^{\frac{1}{2}} \left(\frac{1}{10} + \frac{3y}{5} \right) dy = \frac{13}{160} .
 \end{aligned}$$

Marginal distributions The marginal distribution of X alone and Y alone are

$P_X(x) = \sum_y p(x, y)$ and $P_Y(y) = \sum_x p(x, y)$ if X, Y are discrete random variables

$f_X(x) = \int_{-\infty}^{\infty} f(x, y) dy$ and $f_Y(y) = \int_{-\infty}^{\infty} f(x, y) dx$ if X, Y are continuous random variables

Conditional distributions

Discrete case

The conditional distribution of Y given $X=x$ is

$$P(Y|X) = \frac{p(x,y)}{P_X(x)} \quad \text{provided } p_X(x) > 0$$

The conditional distribution of X given $Y=y$ is

$$P(X|Y) = \frac{p(x,y)}{P_Y(y)} \quad \text{provided } p_Y(y) > 0.$$

Continuous case

The conditional distribution of Y given $X=x$ is

$$f(Y|X) = \frac{f(x,y)}{f_X(x)} \quad \text{provided } f_X(x) > 0$$

The conditional distribution of X given $Y=y$ is

$$f(X|Y) = \frac{f(x,y)}{f_Y(y)} \quad \text{provided } f_Y(y) > 0$$

Remark

$$1) P(a < X < b \mid Y = y) = \begin{cases} \sum_{a < x < b} P(X|Y) & \text{if } X, Y \text{ are discrete} \\ \int_a^b f(x|y) dx & \text{if } X, Y \text{ are continuous.} \end{cases}$$

$$2) P(a < Y < b \mid X = x) = \begin{cases} \sum_{a < y < b} P(Y|X) & \text{for } X, Y \text{ are discrete.} \\ \int_a^b f(y|x) dy & \text{for } X, Y \text{ are continuous.} \end{cases}$$